

First-line talazoparib with enzalutamide in HRR-deficient metastatic castration-resistant prostate cancer: the phase 3 TALAPRO-2 trial

Received: 18 September 2023

Accepted: 10 November 2023

Published online: 04 December 2023

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Preclinical evidence has suggested an interplay between the androgen receptor, which largely drives the growth of prostate cancer cells, and poly(ADP-ribose) polymerase. This association provides a rationale for their co-inhibition for the treatment of metastatic castration-resistant prostate cancer (mCRPC), an area of unmet medical need. The phase 3 TALAPRO-2 study investigated combining the poly(ADP-ribose) polymerase inhibitor talazoparib with enzalutamide versus enzalutamide alone as first-line treatment of mCRPC. Patients were prospectively assessed for tumor alterations in DNA damage response genes involved in homologous recombination repair (HRR). Two cohorts were enrolled sequentially: an all-comers cohort that was enrolled first (cohort 1; $N = 805$ (169 were HRR-deficient)), followed by an HRR-deficient-only cohort (cohort 2; $N = 230$). We present results from the alpha-controlled primary analysis for the combined HRR-deficient population ($N = 399$). Patients were randomized in a 1:1 ratio to talazoparib or placebo, plus enzalutamide. The primary endpoint, radiographic progression-free survival, was met (median not reached at the time of the analysis for the talazoparib group versus 13.8 months for the placebo group; hazard ratio, 0.45; 95% confidence interval, 0.33 to 0.61; $P < 0.0001$). Data for overall survival, a key secondary endpoint, are immature but favor talazoparib (hazard ratio, 0.69; 95% confidence interval, 0.46 to 1.03; $P = 0.07$). Common adverse events in the talazoparib group were anemia, fatigue and neutropenia. Combining talazoparib with enzalutamide significantly improved radiographic progression-free survival in patients with mCRPC harboring HRR gene alterations, supporting talazoparib plus enzalutamide as a potential first-line treatment for these patients. ClinicalTrials.gov Identifier: [NCT03395197](https://clinicaltrials.gov/ct2/show/study/NCT03395197).

Recent approvals of new treatments have led to improved outcomes for patients with advanced prostate cancer^{1,2}. However, metastatic disease remains aggressive and progression is inevitable, necessitating additional therapies for this population of often elderly men^{1,3,4}.

Around a quarter of advanced prostate cancers have alterations in DNA damage response genes involved directly or indirectly in homologous recombination repair (HRR), including *BRCA1/BRCA2* (refs. 5–8); these can sensitize tumors to treatment with poly(ADP-ribose) polymerase

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(PARP) inhibitors^{9–14}. PARP inhibition as monotherapy is an established standard of care for those patients with late-stage prostate cancer.

Preclinical evidence suggests interplay between the androgen receptor, which largely drives the growth of prostate cancer cells, and PARP, providing a rationale for their co-inhibition^{15,16}. Androgen receptor inhibition is associated with upregulated PARP activity and downregulated HRR gene expression^{17,18}, while PARP inhibition suppresses androgen receptor transcriptional activity^{19–21}.

Monotherapy with the PARP inhibitor talazoparib (1 mg per day) showed durable antitumor activity and a favorable benefit–risk profile in patients with heavily pretreated mCRPC with HRR gene alterations in the phase 2, TALAPRO-1 study²². TALAPRO-2 is a multinational phase 3 study evaluating talazoparib in combination with the androgen receptor inhibitor enzalutamide as a first-line treatment in patients with mCRPC²³. An initial, non-randomized, open-label run-in study (part 1; $N = 19$) showed that when combined with enzalutamide at 160 mg per day, similar talazoparib exposure levels to the recommended monotherapy dose were achieved at 0.5 mg per day, establishing this as the starting dose for the combination^{23,24}. Patients were then enrolled sequentially in two cohorts: unselected (cohort 1, all-comers cohort, recruited first) for alterations in DNA damage response genes directly or indirectly involved in HRR and selected (cohort 2) to ensure exclusive enrollment of patients with HRR-deficient disease. The first 805 patients with ($N = 169$) and without ($N = 636$) HRR gene alterations were enrolled as all-comers in cohort 1. Subsequently, an additional 230 patients selected for HRR gene alterations were recruited to complete the predefined enrollment for a combined HRR-deficient population ($N = 399$; Extended Data Fig. 1). All patients were prospectively tested for HRR gene alterations²³.

A recent analysis of the all-comers population of TALAPRO-2 revealed significant improvement in radiographic progression-free survival (rPFS) for talazoparib plus enzalutamide compared with enzalutamide as standard of care (hazard ratio (HR), 0.63; 95% confidence interval (CI), 0.51 to 0.78; $P < 0.0001$)²⁵. Here, we report results of the prespecified alpha-powered independent analysis for the combined HRR-deficient population from both cohorts of TALAPRO-2.

Results

Patients

Between 18 December 2018 and 20 January 2022, 399 patients with HRR gene alterations were enrolled (169 enrolled during the accrual of the all-comers cohort; 230 additional patients to complete the planned accrual target of the combined HRR-deficient population; Fig. 1). Of the 399 enrolled patients, 397 had available prospective tumor tissue test results. Of these, 236 patients with central laboratory, nonhistorical tissue records also had blood samples that underwent concurrent prospective circulating tumor DNA testing after a protocol amendment (26 February 2020). The remaining 2 of 399 patients were enrolled based on circulating tumor DNA testing alone ($n = 1$) or had unspecified tissue source ($n = 1$).

The data cutoff date for the HRR-deficient cohort was 3 October 2022. Baseline characteristics were well balanced (Table 1 and Extended Data Table 1); representativeness of the patients is addressed in Extended Data Table 2. The most commonly altered HRR genes were *BRCA2*, *ATM* and *CDK12*.

Efficacy

Median follow-up for rPFS was 17.5 and 16.8 months for the talazoparib and placebo groups, respectively. Talazoparib plus enzalutamide significantly improved rPFS by blinded independent central review compared with placebo plus enzalutamide (HR, 0.45; 95% CI, 0.33 to 0.61; $P < 0.0001$; median not reached at the time of the analysis versus 13.8 months; Fig. 2).

A consistent treatment effect for rPFS was observed across prespecified clinical subgroups (Fig. 3a) and by investigator assessment

Table 1 | Summary of baseline characteristics (HRR-deficient intention-to-treat population)

Characteristic	Talazoparib + enzalutamide (N=200)	Placebo + enzalutamide (N=199)
Median age (range)—years	70 (41–90)	71 (44–90)
Race		
White	137 (68)	136 (68)
Black or African American	6 (3)	5 (3)
Asian	45 (22)	39 (20)
Multiracial	0	1 (<1)
Other ^a	1 (<1)	1 (<1)
Not reported or unknown	11 (6)	17 (9)
Median baseline serum PSA (range)— $\mu\text{g l}^{-1}$	19.6 (0.2–3412.0)	18.0 (0.0–1055.0)
Gleason score ^b		
<8	42 (21)	52 (26)
≥ 8	152 (76)	143 (72)
Disease site		
Bone (including with soft tissue component)	175 (88)	158 (79)
Lymph node	82 (41)	94 (47)
Visceral (lung)	23 (12)	26 (13)
Visceral (liver)	9 (4)	6 (3)
Other soft tissue	23 (12)	20 (10)
ECOG performance status		
0	128 (64)	118 (59)
1	72 (36)	81 (41)
Prior treatment with a second-generation androgen receptor pathway inhibitor	17 (9)	17 (9)
Abiraterone	16 (8)	16 (8)
Orteronel	1 (<1)	1 (<1)
Prior taxane-based chemotherapy ^c	57 (28)	60 (30)
Patients with at least one alteration in corresponding HRR gene ^d	198 (99)	197 (99)
<i>ATM</i>	47 (24)	39 (20)
<i>ATR</i>	3 (2)	12 (6)
<i>BRCA1</i>	11 (6)	12 (6)
<i>BRCA2</i>	62 (31)	73 (37)
<i>CDK12</i>	36 (18)	39 (20)
<i>CHEK2</i>	34 (17)	37 (19)
<i>FANCA</i>	4 (2)	5 (3)
<i>MLH1</i>	9 (4)	1 (<1)
<i>MRE11A</i>	1 (<1)	2 (1)
<i>NBN</i>	8 (4)	3 (2)
<i>PALB2</i>	9 (4)	8 (4)
<i>RAD51C</i>	2 (1)	2 (1)

Data are n (%), unless otherwise indicated. ^aAmerican Indian, Alaska Native, Native Hawaiian or Other Pacific Islander. ^bNot reported for the remaining patients. ^cAll received docetaxel; HRR-deficient safety population. ^d $N = 3$ patients (1, talazoparib plus enzalutamide; 2, placebo plus enzalutamide) did not have HRR gene alterations and 1 patient in the talazoparib group was of unknown HRR gene alteration status. ECOG, Eastern Cooperative Oncology Group.

(Extended Data Fig. 2). Among 149 patients who had received prior abiraterone or orteronel (a CYP17 inhibitor) or docetaxel for castration-sensitive disease, the HR was 0.43 (95% CI, 0.26 to

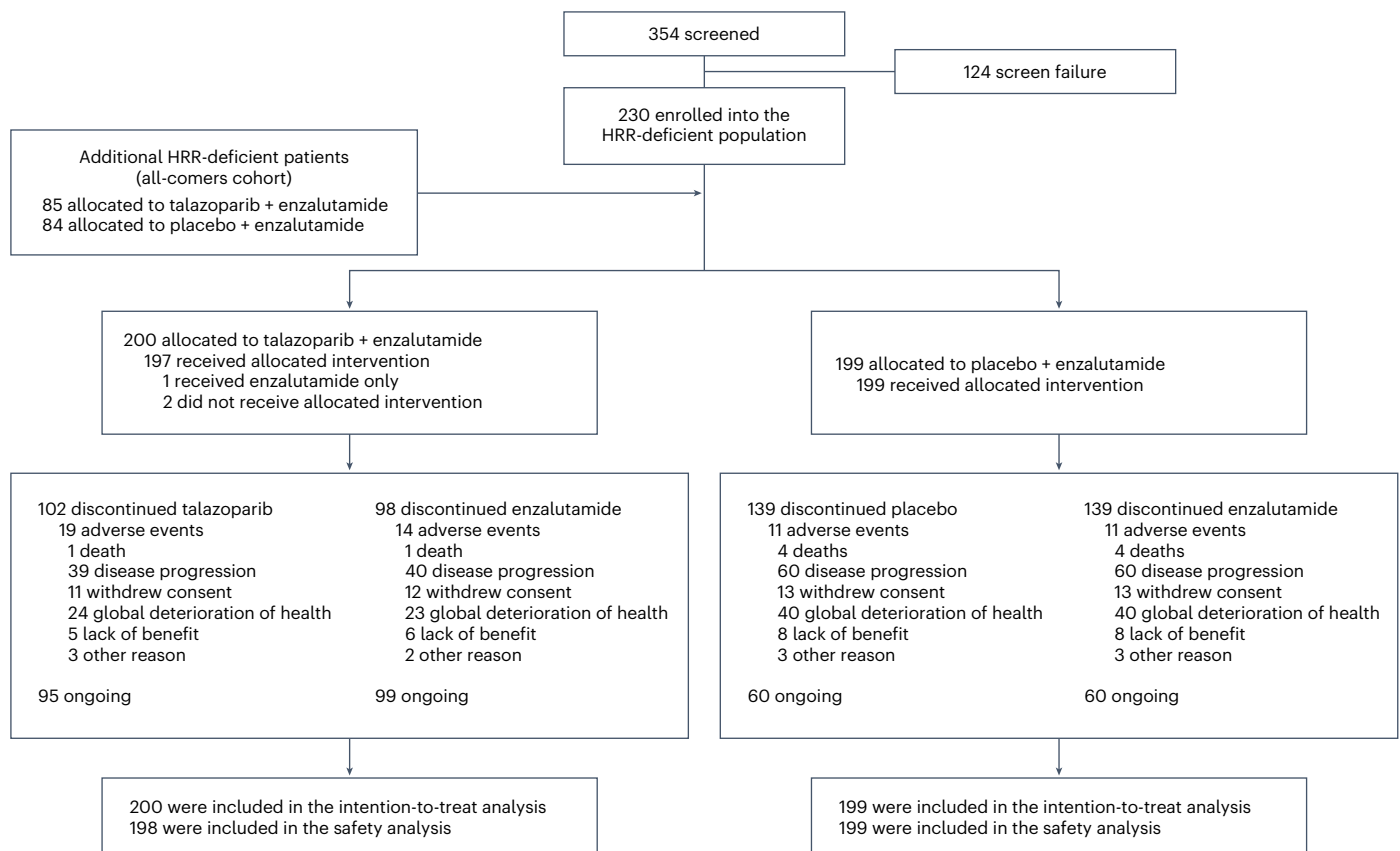


Fig. 1 | Trial profile. Flow diagram showing participant recruitment into the HRR-deficient population, randomization, follow-up and analysis populations.

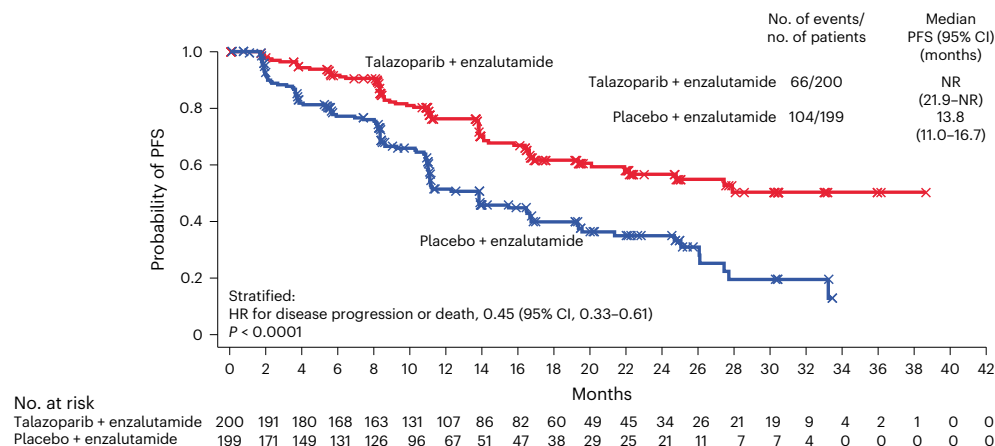


Fig. 2 | rPFS in patients with any HRR gene alteration (assessed by blinded independent central review; HRR-deficient intention-to-treat population). rPFS was compared between treatment groups using stratified log-rank test. HRs and associated 95% two-sided CIs were estimated by a Cox proportional hazards

model. Median time to event was estimated by the Kaplan–Meier method, and 95% CIs were based on the Brookmeyer–Crowley method. The P value is two-sided. NR, not reached at the time of the analysis.

0.70; $P = 0.0006$) in favor of talazoparib plus enzalutamide. Among patients who had received abiraterone or orteronel ($n = 34$), the HR was 0.53 (95% CI, 0.20 to 1.42; $P = 0.20$), and among those who had received docetaxel ($n = 117$), the HR was 0.39 (95% CI, 0.22 to 0.69; $P = 0.0008$).

In a post hoc analysis, patients with *BRCA1/BRCA2* alterations had an 80% lower risk of radiographic progression or death (HR, 0.20; 95% CI, 0.11 to 0.36; $P < 0.0001$; Fig. 3a); those with non-*BRCA1/BRCA2* alterations had a 32% lower risk (HR, 0.68; 95% CI, 0.46 to 1.02; $P = 0.06$)

with talazoparib plus enzalutamide. Further, notable improvements in rPFS were observed with talazoparib plus enzalutamide in the *BRCA2* single-gene subgroup, and in *BRCA* and *CDK12* clusters (Fig. 3b).

Overall survival data remain immature, with the majority of patients being alive: 43 (22%) patients in the talazoparib group and 53 (27%) in the placebo group had died at data cutoff. Three patients in the talazoparib group and 18 in the placebo group subsequently received a PARP inhibitor (all received olaparib) per the treating physician's judgment and local approval and availability of a PARP inhibitor.

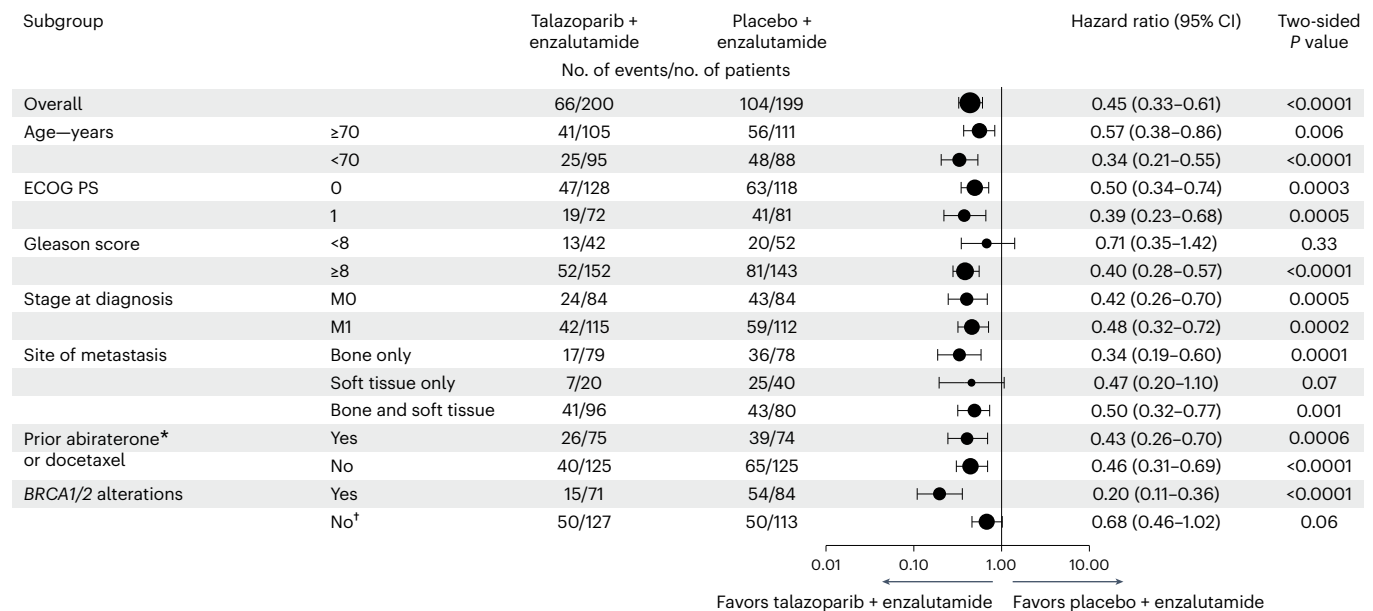
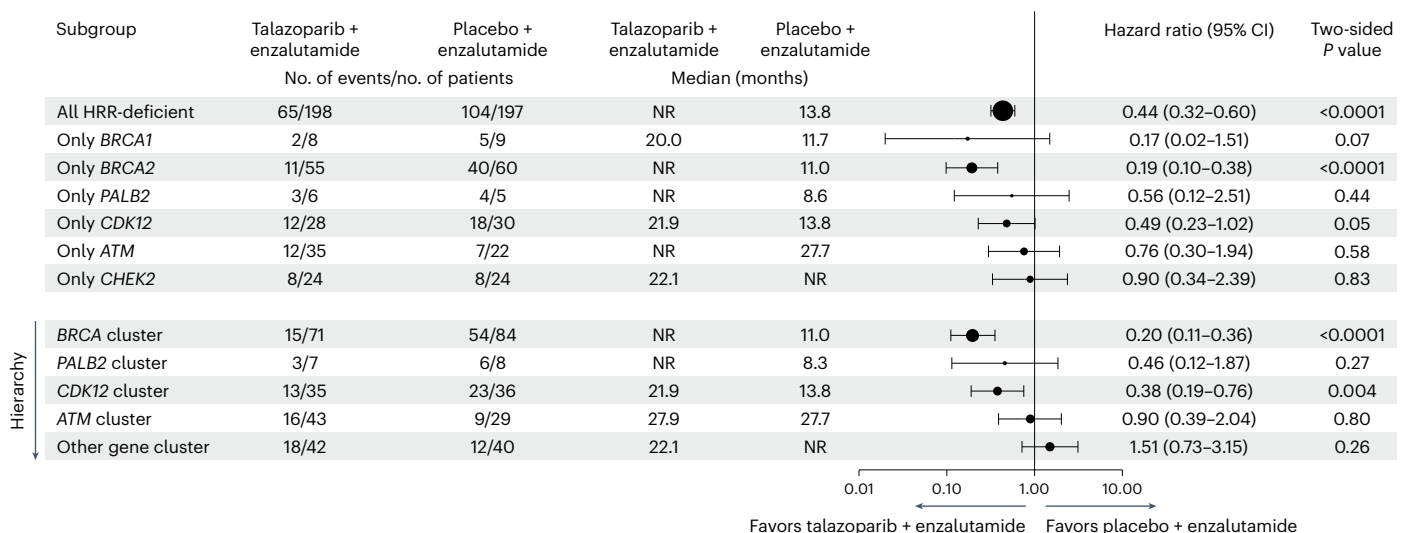
a By baseline characteristics**b** By selected gene subgroups*

Fig. 3 | Subgroup analysis of rPFS. a, b. Subgroup analysis of rPFS by baseline characteristics (**a**) and by gene subgroups (**b**) (assessed by blinded independent central review; HRR-deficient intention-to-treat population). The overall HR for all patients, and by *BRCA1/BRCA2* alteration status, was based on a Cox proportional hazards model stratified by the randomization stratification factors. For all other subgroups, the HR was based on an unstratified Cox model with treatment as the only covariate. Data are presented as HRs with two-sided 95% CIs. P values are two sided. The asterisk indicates the inclusion of one patient in each treatment arm who received prior orteronel. †Excludes four patients who did not have HRR gene alterations but were incorrectly randomized to the

HRR-deficient population; including these patients resulted in an HR of 0.72 (95% CI, 0.49 to 1.07) for the non-*BRCA* alterations subgroup. ‡Post hoc exploratory analysis; as this analysis was underpowered, the data are hypothesis-generating and should be interpreted with caution. Gene clustering alteration dominance hierarchy is any *BRCA1/BRCA2* alteration (*BRCA* cluster), then any *PALB2* (*PALB2* cluster), next any *CDK12* (*CDK12* cluster), then any *ATM* (*ATM* cluster), and finally, any of all other genes (with each patient counted only once). For the single-gene subgroups, only patients bearing alteration(s) in the designated HRR gene and none of the other HRR genes tested are shown, with a prevalence cutoff for display of ≥10 across arms. PS, performance status.

The HR for death was 0.69 (95% CI, 0.46 to 1.03; $P = 0.07$; Fig. 4a). In the *BRCA1/BRCA2* and non-*BRCA1/BRCA2* altered subgroups, the HRs for death were 0.61 (95% CI, 0.31 to 1.23; $P = 0.16$) and 0.66 (95% CI, 0.40 to 1.10; $P = 0.11$), respectively.

Confirmed objective response rate in patients with measurable disease at baseline was 67% (49/73; 95% CI, 55.1% to 77.7%) for the talazoparib group and 40% (26/65; 95% CI, 28.0% to 52.9%) for the placebo group (Fig. 4b). Time to prostate-specific antigen (PSA) progression, time to initiation of cytotoxic chemotherapy and investigator-assessed time to progression or death on first subsequent antineoplastic

therapy were significantly prolonged in the talazoparib group (Fig. 4c–e; see Extended Data Table 3 for results of other secondary efficacy endpoints).

Safety

Median duration of treatment was 14.6 months for talazoparib and 14.7 months for enzalutamide in the talazoparib group, and 12.0 months for placebo and 12.1 months for enzalutamide in the placebo group. Median relative dose intensities in the talazoparib group were 81% for talazoparib and 100% for enzalutamide; 10% of the talazoparib group

had moderate renal impairment at baseline requiring a starting dose of talazoparib of 0.35 mg per day.

The most common adverse events in the talazoparib group were anemia, fatigue, neutropenia, thrombocytopenia, nausea and decreased appetite. In the placebo group, fatigue, back pain and arthralgia were the most common adverse events (Table 2). The most common grade ≥ 3 adverse event in the talazoparib group was anemia (41%; Table 2), with a median time to onset of 3.3 months, and requiring dose modification of talazoparib according to the protocol. Thirty-six percent of patients in the talazoparib group received a packed red blood cell transfusion. At baseline, 56% of patients in the talazoparib group had grade 1–2 anemia. Only 4% of patients in the talazoparib group discontinued talazoparib due to anemia.

There were more dose interruptions and reductions due to adverse events in the talazoparib group than in the placebo group, but permanent discontinuation rates were similar (discontinuation of talazoparib in 10% versus placebo in 7% of patients; discontinuation of enzalutamide in 8% versus 7%; Table 2).

After a median follow-up for safety of 15.4 and 12.9 months for the talazoparib and placebo groups, respectively, no cases of myelodysplastic syndrome or acute myeloid leukemia were reported. Venous embolic and thrombotic events were reported in seven patients in the talazoparib group and two patients in the placebo group. There were four cases of pulmonary embolism (one grade 2, three grade 3) in the talazoparib group and two cases (both grade 3) in the placebo group. There were no treatment-related deaths.

Discussion

In one of the largest studies in patients with mCRPC with HRR gene alterations, the prospectively defined, alpha-controlled primary outcome in the combined HRR-deficient population showed that talazoparib plus enzalutamide resulted in a clinically meaningful and statistically significant 55% reduction in risk of progression or death versus placebo plus enzalutamide as first-line treatment. These results build on the previous subgroup analysis of the HRR-deficient patients in the all-comers cohort, which showed a 54% reduction in risk of progression or death for talazoparib plus enzalutamide versus placebo plus enzalutamide (HR, 0.46; 95% CI, 0.30 to 0.70; $P = 0.0003$)²⁵. Although overall survival data are immature and statistical significance was not reached, interim data favor this combination. Other key secondary endpoints, including time to PSA progression, time to cytotoxic chemotherapy and time to progression or death on the first subsequent antineoplastic therapy, favored the talazoparib group.

Importantly, TALAPRO-2 was not enriched for patients with *BRCA1/BRCA2* alterations, which were well balanced between the treatment arms (talazoparib group, 36%; placebo group, 42%); the observed *BRCA1/BRCA2* prevalence in the prospectively determined HRR-deficient population was in line with previous reports^{5,7,8}. This is notable since *BRCA* alterations are a strong predictive factor toward improved treatment outcomes for patients receiving PARP inhibitor monotherapy²⁶. Talazoparib plus enzalutamide reduced risk of

progression or death by 80% in the *BRCA1/BRCA2* subgroup and by 32% in the non-*BRCA1/BRCA2* subgroup. The *CDK12* results are striking given an alteration prevalence of 5% to 7% according to the literature and that limited clinical data indicate poor prognosis with minimal benefit from PARP inhibitor monotherapy in patients who have prostate cancer and *CDK12* alterations²⁷. *CDK12* deficiency is associated with a

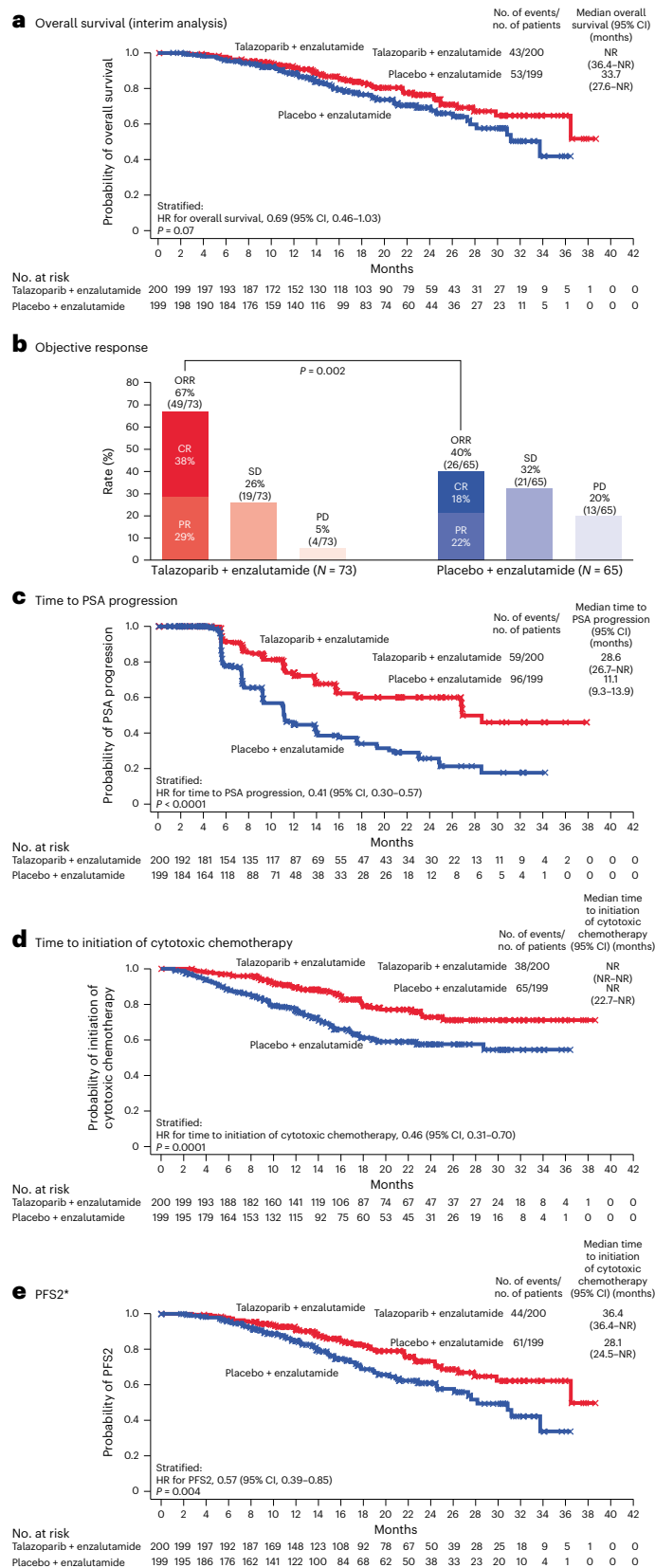


Fig. 4 | Secondary efficacy endpoints. a–e, Secondary efficacy endpoints: overall survival (a), objective response (b), time to PSA progression (c), time to initiation of cytotoxic chemotherapy (d) and PFS2 (e) (HRR-deficient intention-to-treat population). Time-to-event endpoints were compared between treatment groups using a stratified log-rank test. HRs and associated 95% two-sided CIs were estimated by a Cox proportional hazards model. Median time-to-event endpoints were estimated by the Kaplan–Meier method, and 95% CIs were based on the Brookmeyer–Crowley method. P values are two sided. The asterisk denotes that PFS2 was based on investigator assessment (time from randomization to the date of documented progression on the first subsequent antineoplastic therapy or death from any cause, whichever occurs first). CR, complete response; ORR, objective response rate; PD, progressive disease; PFS2, progression-free survival 2; PR, partial response; SD, stable disease.

Table 2 | Summary of treatment-emergent adverse events (HRR-deficient safety population)^a

Adverse event	Talazoparib+ enzalutamide (N=198)		Placebo+ enzalutamide (N=199)	
	All grades	Grade ≥3	All grades	Grade ≥3
Any adverse event	196 (99)	134 (68)	191 (96)	79 (40)
Treatment-related adverse event	180 (91)	105 (53)	144 (72)	28 (14)
Serious adverse event	60 (30)	54 (27)	40 (20)	32 (16)
Serious and treatment-related adverse event	27 (14)	23 (12)	0	0
Adverse event resulting in dose interruption of:				
Talazoparib/placebo ^b	114 (58)	•	34 (17)	•
Enzalutamide ^c	67 (34)	•	31 (16)	•
Adverse event resulting in dose reduction of:				
Talazoparib/placebo ^b	103 (52)	•	11 (6)	•
Enzalutamide ^c	28 (14)	•	12 (6)	•
Adverse event resulting in permanent drug discontinuation of:				
Talazoparib/placebo ^b	20 (10)	•	14 (7)	•
Enzalutamide ^c	15 (8)	•	14 (7)	•
Grade 5 adverse event	3 (2) ^d	•	5 (3) ^d	•
Most common adverse events (all grades in ≥10% of patients) ^e				
Anemia	128 (65)	81 (41)	31 (16)	9 (5)
Fatigue	66 (33)	3 (2)	53 (27)	2 (1)
Neutropenia	64 (32)	37 (19)	13 (7)	2 (1)
Thrombocytopenia	49 (25)	14 (7)	5 (3)	1 (<1)
Nausea	41 (21)	3 (2)	34 (17)	1 (<1)
Decreased appetite	40 (20)	2 (1)	28 (14)	2 (1)
Back pain	39 (20)	3 (2)	44 (22)	2 (1)
Leukopenia	37 (19)	11 (6)	15 (8)	0
Hypertension	36 (18)	16 (8)	38 (19)	16 (8)
Asthenia	31 (16)	4 (2)	29 (15)	0
Constipation	26 (13)	0	33 (17)	0
Fall	26 (13)	4 (2)	24 (12)	3 (2)
Arthralgia	25 (13)	0	44 (22)	0
Diarrhea	24 (12)	0	22 (11)	0
Hot flush	23 (12)	0	28 (14)	0
Dizziness	20 (10)	1 (<1)	15 (8)	2 (1)
Headache	12 (6)	0	22 (11)	1 (<1)

Data are n (%). ^aShown are adverse events that occurred from the time of the first dose of study treatment through 28 d after permanent discontinuation of all study treatments or before initiation of a new antineoplastic or any investigational therapy, whichever occurs first. Adverse events were graded according to National Cancer Institute Common Terminology Criteria for Adverse Events version 4.03. All data are reported per the safety population defined as all patients who were treated with at least one dose of study treatment, including one patient who was randomized to talazoparib plus enzalutamide but received enzalutamide only (patients treated with both study treatments: N=197 for talazoparib plus enzalutamide; N=199 for placebo plus enzalutamide). ^bIncludes permanent discontinuation/dose reduction/dose interruption of talazoparib/placebo only plus permanent discontinuation/dose reduction/dose interruption of both talazoparib/placebo and enzalutamide. ^cIncludes permanent discontinuation/dose reduction/dose interruption of enzalutamide only plus permanent discontinuation/dose reduction/dose interruption of both talazoparib/placebo and enzalutamide. ^dNone were considered treatment related. ^eNone of these events were recorded as grade 5.

distinct chromosomal damage signature and disrupted replication and transcription^{28,29}, perhaps resulting in vulnerability to the combination of enzalutamide and talazoparib, a potent PARP trapper³⁰. Although patient numbers were very small and the CIs wide, a similar benefit to that seen with *BRCA2* was also apparent in the *BRCA1* single-gene subgroup, with a smaller benefit apparent in the *PALB2* cluster. However, these post hoc analyses were underpowered and hypothesis generating, so the results should be interpreted with caution.

Two other recent phase 3 trials have explored the combination of PARP inhibitors and the androgen biosynthesis inhibitor abiraterone acetate/prednisone as first-line treatment for mCRPC. The PROpel (NCT03732820) trial, which enrolled all-comers without prospective assessment of *BRCA* or HRR status, demonstrated improved rPFS with the combination of olaparib plus abiraterone versus placebo plus abiraterone for patients with HRR gene alterations (HR, 0.50; 95% CI, 0.34 to 0.73)³¹. Exploratory analysis in the *BRCA1/BRCA2* subgroup showed an HR of 0.24 (95% CI, 0.12 to 0.45)³². The MAGNITUDE (NCT03748641) trial also showed improved rPFS with the combination of niraparib plus abiraterone versus placebo plus abiraterone for patients with HRR gene alterations (HR, 0.73; 95% CI, 0.56 to 0.96), with particular benefit in the *BRCA1/BRCA2* subgroup (HR, 0.53; 95% CI, 0.36 to 0.79)³³. Exploratory single-gene analysis of the MAGNITUDE trial, although underpowered, showed potential benefit of combined PARP and androgen receptor inhibition in patients with a Fanconi anemia pathway gene alteration (*PALB2*, *BRIPI* and *FANCA*) beyond *BRCA1/BRCA2* (ref. 34), whereas a lack of differential benefit was seen in tumors with *CDK12* alterations³⁴. Results from the PROfound (NCT02987543)¹³ and TRITON-3 (NCT02975934)¹⁴ phase 3 trials of PARP monotherapy (olaparib and rucaparib, respectively) in patients with pretreated mCRPC also indicated that patients with *BRCA2* alterations derived benefit. There was inconclusive evidence supporting *BRCA1* due to small patient numbers, preliminary positive evidence for *CDK12* in PROfound (HR below 1 but wide CIs) and lack of efficacy with *ATM*^{13,14}.

The main limitations of this study are due to the rapidly changing treatment landscape for patients with mCRPC. For example, the use of PARP inhibitors as a subsequent therapy was limited to a small number of patients (3 in the talazoparib group and 18 in the placebo group; all received olaparib). This small number most likely reflects the limited availability of PARP inhibitors for the treatment of mCRPC when the TALAPRO-2 study was carried out. Based on established phase 3 data³⁵, it is anticipated that survival in the placebo group in those who did not receive a subsequent PARP inhibitor may be shorter than in those who did. Also, the use of androgen receptor-targeted therapy has become more commonplace since patients were recruited to the TALAPRO-2 study^{36–38}. Over one-third of the HRR-deficient population in TALAPRO-2 had received prior docetaxel or abiraterone for castration-sensitive disease³⁶, and these patients had a significant 57% reduction in risk of radiographic progression or death. However, only 8% of patients in either arm had received prior abiraterone; the benefit in these patients is hypothesis generating and warrants further studies.

The safety profile of talazoparib plus enzalutamide was closely aligned with that observed in the previously reported all-comers population²⁵. The incidence of anemia, including grade 3 and 4 events, was higher than with talazoparib monotherapy^{22,39}. Anemia was managed through close patient monitoring, protocol-mandated dose interruption to permit recovery followed by dose reduction for grade ≥3 anemia (once hemoglobin levels were <8 g dL⁻¹; to optimize individual treatment), and supportive measures, including packed red blood cell transfusions. To reflect the real-world patient population of mCRPC, often with bone metastases and bone marrow insufficiency, TALAPRO-2 could enroll patients with hemoglobin levels as low as 9 g dL⁻¹. Notably, more than half of the patients (56%) had grade 1 and 2 anemia at baseline. Although 41% developed grade 3 and 4 anemia after a median talazoparib treatment duration of 3.3 months, only 4% of patients discontinued talazoparib because of anemia. Importantly, no cases of myelodysplastic syndrome

or acute myeloid leukemia were reported. The incidence of permanent discontinuation was similar between the treatment groups, and the median relative dose intensity of talazoparib remained high at >80%.

In conclusion, these results support the use of talazoparib plus enzalutamide as a potential first-line treatment option for patients with mCRPC harboring tumor HRR gene alterations.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41591-023-02704-x>.

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Methods

Trial design and patients

TALAPRO-2 (NCT03395197) is an ongoing double-blind, randomized, placebo-controlled trial. Details of the trial design have been published²³ and are in the protocol (Supplementary Protocol).

Eligibility criteria included ongoing androgen deprivation therapy; asymptomatic or mildly symptomatic mCRPC with HRR gene alterations; Eastern Cooperative Oncology Group performance status score of 0 or 1; progressive disease; adequate bone marrow function (hemoglobin ≥ 9 g dl⁻¹); and no prior life-prolonging systemic therapy for castration-resistant disease²³. Prior docetaxel and abiraterone or orteronel in the castration-sensitive setting were allowed. Patients were randomized in a 1:1 ratio (using a centralized, interactive web response system and a permuted block size of 4) to receive 0.5 mg talazoparib (moderate renal impairment, 0.35 mg) or placebo (all received enzalutamide, 160 mg) once daily. Randomization was stratified by prior second-generation androgen receptor pathway inhibitor (abiraterone or orteronel) or docetaxel (yes/no). Formal crossover from the placebo group to the talazoparib group was not part of the study design.

Before randomization, patients consented to provide solid tumor tissue (de novo or archival) and/or blood-based samples, for prospective assessment of HRR gene alterations (*BRCA1*, *BRCA2*, *PALB2*, *ATM*, *ATR*, *CHEK2*, *FANCA*, *RAD51C*, *NBN*, *MLH1*, *MRE11A*, *CDK12*) using FoundationOne® CDx and/or FoundationOne Liquid® CDx (Foundation Medicine). Historical FoundationOne® test results could also be used. Patients were considered HRR-deficient if they had one or more alteration(s) in at least one of these 12 genes. For prospective HRR status determination, test records generated after the randomization date were excluded. Alterations were defined as truncating short variants, selected inactivating short variants identified as known/likely pathogenic per FoundationOne® pipeline, inactivating rearrangements or homozygous deletion of one or more exons. The definition of HRR alterations was the same for FoundationOne® Liquid CDx, except homozygous deletion of one or more exons was limited to *BRCA1*/*BRCA2* only. Notably, patients with heterozygous deletions of one or more exons alone were not enrolled in the HRR-deficient population.

Enrollment of patients with *ATM* and/or *CDK12* gene alterations was paused between January and November 2021 as their observed prevalence exceeded expectations⁷ and was anticipated to suppress the representation of alterations in the remaining genes under study. The pause in enrollment of patients with *ATM* and/or *CDK12* gene alterations was driven by expected prevalence numbers based on the largest and most comprehensive prospective assessment of prostate cancer tumor samples using the FoundationOne® assay⁷. This pause occurred in a blinded fashion regarding distribution of HRR alterations to the two treatment arms and allowed a rebalancing of the distribution across the 12-gene panel in an effort to best reflect the prevalence in mCRPC⁷.

Study treatment continued until radiographic progression, adverse event leading to permanent discontinuation, patient decision to discontinue or death. Treatment could continue after radiographic progression if the investigator determined benefit was still being derived.

The trial was conducted in accordance with the International Ethical Guidelines for Biomedical Research Involving Human Subjects, Good Clinical Practice guidelines, the principles of the Declaration of Helsinki and local laws. The protocol and amendments were approved by the institutional review board and independent ethics committee for each site. The following independent ethics committees or Institutional Review Boards provided study approval: Comité de Revisión Institucional - Hospital Británico de Buenos Aires, CABA, Argentina; Comité de Ética 'Dr. Claude Bernard', Rosario, Argentina; Comité de Ética en Investigación - Centro de Educación Médica e Investigaciones Clínicas 'Norberto Quirno' - CEMIC, CABA, Argentina; Comité de Ética en Investigación de la Fundación OncoSalud (CEIFOS), Pergamino, Argentina; Comité Independiente de Ética Para Ensayos En

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of Korea; Health and Disability Ethics Committee, Wellington, New Zealand; REK Sor-Ost, Oslo, Norway; Comité Institucional de Ética en Investigación del INEN, Lima, Peru; Comité Institucional de Bioética de Via Libre, Lima, Peru; Komisja Bioetyczna przy Okręgowej Izbie Lekarskiej w Gdańsku, Gdańsk, Poland; Comissão de Ética para a Investigação Clínica, Lisboa, Portugal; University of the Witwatersrand Human Research Ethics Committee (Medical), Johannesburg, South Africa; CEIm del Hospital Universitari Vall d'Hebron, Barcelona, Spain; Etikprovningsmyndigheten, Uppsala, Sweden; Health and Care Research Wales, Wales REC 3, Cardiff, United Kingdom; Advarra Institutional Review Board, Columbia, MD, United States; Vanderbilt Human Research Protection Program (VHRPP) Institutional Review Board, Nashville, TN, United States; University of Utah Institutional Review Board, Salt Lake City, UT, United States; Biomedical Research Alliance of New York, LLC/Institutional Review Board, Lake Success, NY, United States; Western Institutional Review Board, Puyallup, WA, United States; Sharp HealthCare Institutional Review Board, San Diego, CA, United States; Schulman Associates Institutional Review Board, Cincinnati, OH, United States; Loma Linda University Health - Institutional Review Board, Loma Linda, CA, United States; Administrative Panels on Human Subjects in Medical Research ('Stanford Institutional Review Board'), Palo Alto, CA, United States; University of Maryland, Baltimore - Institutional Review Board, Baltimore, MD, United States; Cook County Health Office of Research and Regulatory Affairs, Chicago, IL, United States; Samaritan Health Services Regional Institutional Review Board, Corvallis, OR, United States; University of Iowa Institutional Review Board-01, Human Subjects Office, Iowa City, IA, United States; Lakeland Regional Medical Center, Inc. Institutional Review Board, Lakeland, FL, United States; VA Med Ctr, Long Beach CA Institutional Review Board #1, Long Beach, CA, United States; Rush University Medical Center Institutional Review Board, Chicago, IL, United States; UCLA Office of the Human Research Protection Program, Los Angeles, CA, United States; VA Saint Louis Healthcare System Institutional Review Board, St. Louis, MO, United States; Baylor Scott and White Research Institute Institutional Review Board-Gold, Temple, TX, United States; Providence St. Joseph Health Institutional Review Board, Renton, WA, United States; IntegReview, Austin, TX, United States; Kaiser Permanente Northwest Institutional Review Board, Portland, OR, United States; Ochsner Institutional Review Board, New Orleans, LA, United States; Eisenhower Medical Center, Institutional Review Board, Rancho Mirage, CA, United States. All patients provided written informed consent.

Trial endpoints

The primary endpoint was rPFS by blinded independent central review per Response Evaluation Criteria in Solid Tumors (version 1.1; soft tissue disease) and Prostate Cancer Clinical Trials Working Group 3 (bone disease)²³. A full list of secondary endpoints is included in the Supplementary Methods, and these endpoints have been previously listed²³. Planned secondary endpoints not reported in this article are: time to opiate use for prostate cancer pain, pharmacokinetics and patient-reported outcomes.

Exploratory subgroup analyses were conducted for rPFS by baseline characteristics (Supplementary Methods). A post hoc analysis of rPFS by *BRCA1/BRCA2* alteration status (yes/no) and by single genes and hierarchical gene clusters (*BRCA*, *PALB2*, *CDK12*, *ATM* and any of all other HRR genes) was also performed (Supplementary Methods).

Statistical analysis

Approximately 380 patients with HRR gene alterations were to be enrolled. To maintain overall type I error at or below a one-sided alpha level of 0.025, alpha was split equally between the all-comers cohort and HRR-deficient population.

For the primary comparison in the HRR-deficient population, 224 PFS events based on a Lan DeMets α -spending function would provide

85% power to detect an HR of 0.64 using a one-sided stratified log-rank test at a significance level of 0.0125. A prespecified interim analysis of PFS was planned after approximately 70% of the expected events (157 events). The HRR-deficient cohort would be stopped if the efficacy boundary was crossed and an interim efficacy analysis of overall survival would be performed. As the efficacy boundary ($P \leq 0.0038$) was crossed at this interim analysis, this became the final analysis. Other endpoints had no adjustment for multiplicity. Survival and safety follow-up continue.

Time-to-event endpoints were compared between treatment groups using a stratified log-rank test unless otherwise stated. HRs and associated 95% two-sided CIs were estimated by a Cox proportional hazards model. Median time-to-event endpoints were estimated by the Kaplan–Meier method, and 95% CIs based on the Brookmeyer–Crowley method. For subgroup analysis of rPFS (except by *BRCA* status), the HR was based on an unstratified Cox model with treatment as the only covariate due to small patient numbers in some subgroups. Missing/partial dates were imputed as specified per protocol. Other missing data were not imputed. Reported *P* values are two sided.

Oracle Clinical Remote Data Capture was used for data collection, and SAS version 9.4 was used for data analysis.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Upon request, and subject to review, Pfizer will provide the data that support the findings of this study. Subject to certain criteria, conditions and exceptions, Pfizer may also provide access to the related individual de-identified participant data. See <https://www.pfizer.com/science/clinical-trials/trial-data-and-results/> for more information.

Acknowledgements

We thank the patients and their families and caregivers for their participation, as well as the trial centers that supported this trial (trial investigators are listed in the Supplementary Information). This study was sponsored by Pfizer Inc. Astellas Pharma Inc. provided enzalutamide. The authors wish to thank L. Yu and the Pfizer Clinical Programming team. Editorial and medical writing support was provided by A. Smith and E. Messina on behalf of CMC Affinity, a division of IPG Health Medical Communications, and was funded by Pfizer.

Author contributions

K.F., X.L., C.G.H., N.D.i.S., A.D.L., F.Z. and N.A. conceived and designed the study. X.L., C.G.H., N.D.i.S., A.D.L. and F.Z. acquired and analyzed the data. All authors contributed to data interpretation as well as development, writing and approval of the manuscript.

Competing interests

A.A.A. reports honoraria from Aculeus Therapeutics, Amgen, Astellas Pharma, AstraZeneca, Bayer, Bristol Myers Squibb, Daiichi Sankyo, Ipsen, Janssen, Merck Serono, Merck Sharp & Dohme, Novartis, Noxopharm, Pfizer, Sanofi, Telix Pharmaceuticals and Tolmar; consulting fees from Aculeus Therapeutics, Astellas Pharma, Janssen and Novartis; participation on advisory boards for Amgen, Arvinas, Astellas Pharma, AstraZeneca, Bayer, Bristol Myers Squibb, Daiichi Sankyo, Ipsen, Janssen, Merck Serono, Merck Sharp & Dohme, Novartis, Noxopharm, Pfizer, Sanofi, Telix and Tolmar; participation on a data safety monitoring board for OncoSec; research funding (institution unless stated otherwise) from Aptevo Therapeutics, Astellas Pharma (investigator), AstraZeneca (investigator), Bionomics, Bristol Myers Squibb, Exelixis, Gilead Sciences, GlaxoSmithKline, Hinovra Pharmaceuticals, Ipsen, Janssen, Lilly, MedImmune, Merck

Serono (investigator), MSD, Novartis, Pfizer, Sanofi and Synthorx; and travel, accommodations and/or expenses from Amgen, Astellas Pharma, Janssen, Merck Serono, Novartis, Pfizer and Tolmar; and receiving medical writing services from Astellas Pharma, Exelixis and Pfizer; he is Chair of the Urologic Oncology Group for the Clinical Oncology Society of Australia, and Chair of the Translational Research Subcommittee and on the Scientific Advisory Committee for the ANZUP Cancer Trials Group. N.M. reports honoraria (personal) from Sanofi; research funding (institution) from Amgen, Astellas Pharma, AstraZeneca, Bayer, Chugai Pharma, Eisai, Janssen, Lilly, MSD, Pfizer, PRA Health Science, Roche, Seagen, Taiho and Takeda; and travel, accommodations and/or expenses (personal) from Pfizer. J.C. reports a consulting or advisory role for Advanced Accelerator Applications/Novartis, Astellas Pharma, AstraZeneca, Bayer, Bristol Myers Squibb, Johnson & Johnson, MSD Oncology, Pfizer, Roche and Sanofi; participation in speakers' bureau for Astellas Pharma, Bayer and Johnson & Johnson; research funding (institution) from AB Science, Aragon Pharmaceuticals, AROG Pharmaceuticals, Astellas Pharma, AstraZeneca AB, AVEO Pharmaceuticals, Bayer AG, Blueprint Medicines, BN ImmunoTherapeutics, Boehringer Ingelheim España SA, Bristol Myers Squibb International Corporation, Clovis Oncology, Cougar Biotechnology, Deciphera, Exelixis, Genentech, GlaxoSmithKline, Incyte, Janssen-Cilag International NV, Karyopharm Therapeutics, Laboratoires Leurquin Mediolanum, Lilly, MedImmune, Millennium Pharmaceuticals, Nanobiotix, Novartis Farmacéutica SA, Pfizer, Puma Biotechnology, Roche, Sanofi Aventis GmbH, SFJ Pharmaceuticals Group and Teva; and travel, accommodations and/or expenses from AstraZeneca, BMS, Ipsen and Roche. A.P.F. reports honoraria from Astellas Pharma, AstraZeneca, Bristol Myers Squibb, Ipsen, Janssen, MSD, Novartis, Pfizer and Roche; a consulting or advisory role for Bayer, Ipsen, Janssen, MSD, Novartis, Pfizer and Roche; stock or stock options in Brazilian Information Oncology; and research funding from AstraZeneca, Bristol Myers Squibb, CAPES – CNPq, Foundation Medicine, Ipsen, MSD and Roche; and travel, accommodations and/or expenses from Astellas Pharma, AstraZeneca, BMS, Ipsen, Janssen, MSD, Novartis, Pfizer and Roche. U.D.G. reports a consulting or advisory role for Amgen, Astellas Pharma, AstraZeneca, Bayer, Bristol Myers Squibb, Dompé Farmaceutici, Eisai, Ipsen, Janssen, Merck KGaA, MSD, Novartis and Pfizer; research funding (institution) from AstraZeneca, Roche, and Sanofi; and travel, accommodations and/or expenses from AstraZeneca, Ipsen and Pfizer. J.Y.J. declares no competing interests. P.C.C.F. reports a consulting or advisory role for MSD and travel, accommodations and/or expenses from Pfizer. E.V. declares no competing interests. R.J.J. reports honoraria from Astellas Pharma, Bayer, Bristol Myers Squibb, Ipsen, Janssen, Merck Serono, MSD, Pfizer and Roche; a consulting or advisory role for Astellas Pharma, Bayer, Bristol Myers Squibb, Ipsen, Janssen, Merck Serono, MSD, Novartis, Pfizer and Roche; research funding from Astellas Pharma, Bayer, Clovis Oncology, Exelixis and Roche; and travel, accommodations and/or expenses from Bayer and Janssen. N.D.S. reports a consulting or advisory role for AbbVie, Alesia Therapeutics, Akido, Amgen, Arquer, Asieris, Astellas Pharma, AstraZeneca, Bayer, Boston Scientific, Bristol Myers Squibb, CG Oncology, Clarity Pharmaceuticals, Clovis Oncology, Dendreon, Exact Imaging, Exact Sciences, FerGene, Ferring, FIZE Medical, Foundation Medicine, GenesisCare, Genentech, Guardant Health, ImmunityBio, Incyte, Invitae, Janssen, Lantheus, Lilly, Mdxhealth, Merck, Minomic, Myovant Sciences, Myriad Genetics, Nymox, Pacific Edge Biotechnology, Pfizer, Photocure, PlatformQ, Profound, Promaxo, Propella Therapeutics, Protara, Sanofi, Sesen Bio, Speciality Networks, Telix Pharmaceuticals, Tolmar, UroGen Pharma, Vaxiion and Vessi; providing expert testimony for Ferring; and leadership or other fiduciary role in another board, society, committee, or advocacy group with Photocure. C.D. reports participation on advisory boards

for Astellas Pharma, Bayer, Janssen and Pfizer; and research funding from AstraZeneca, Bayer, Dendreon, Hengrui Pharmaceuticals, Janssen, Laekna Therapeutics, Myovant Sciences and Pfizer. S.Z. reports payment or honoraria for lectures, presentations, speakers bureaus, manuscript writing or educational events from Amgen (personal and institution), Astellas (personal and institution), Bayer (personal and institution), Bristol Myers Squibb (personal and institution), Eisai (personal), Janssen (personal), Merck Serono (personal and institution), MSD (institution), Novartis (personal), and Pfizer (personal and institution); participation on a data safety monitoring board or advisory board for Amgen (personal and institution), Bayer (personal and institution), Bristol Myers Squibb (institution), Eisai (personal), Gilead (personal), Ipsen (personal), Janssen (personal), Merck Serono (personal and institution), MSD (institution), Novartis (personal) and Pfizer (institution); research funding (institution) from Eisai; and travel, accommodations and/or expenses from Amgen, Astellas Pharma, AstraZeneca, Bayer, Ipsen, Janssen, Merck Serono, MSD and Pfizer. J.O. reports payment or honoraria for lectures, presentations, speakers bureaus, manuscript writing or educational events from Astellas Pharma, AstraZeneca, Bayer, BMS Norway, Eisai, Ipsen, Janssen-Cilag, Merck and Roche; participation on a data safety monitoring board or advisory board for Astellas Pharma, AstraZeneca, Bayer, BMS, Eisai, Ipsen, Janssen-Cilag, Merck and Roche; and travel, accommodations and/or expenses from Astellas Pharma. D.Y. declares no competing interests. X.L., C.G.H., N.D.i.S., A.D.L. and F.Z. are employees of Pfizer and may hold Pfizer stock/stock options. N.A. has received an honorarium for consultancy before May 2021 from the following: Astellas Pharma, AstraZeneca, AVEO, Bayer, Bristol Myers Squibb, Calithera Biosciences, Eisai, EMD Serono, Exelixis, Foundation Medicine, Genentech, Gilead Sciences, Immunomedics, Janssen, Lilly and MEI Pharma; and research funding

(institution) from Arvinas, Astellas Pharma, AstraZeneca, Bayer, Bristol Myers Squibb, Calithera Biosciences, Celldex, Clovis Oncology, CRISPR Therapeutics, Eisai, EMD Serono, Exelixis, Genentech, Gilead Sciences, GlaxoSmithKline, Immunomedics, Janssen, Lava, Lilly, Merck, Nektar, Neoleukin, Novartis, ORIC Pharmaceuticals, Pfizer, Rexahn, Roche, Sanofi, Seagen, Takeda and TRACON. K.F. reports honoraria (institution) for participation in advisory boards and talks from Advanced Accelerator Applications/Novartis, Amgen, Astellas Pharma, AstraZeneca, Bayer, Clovis Oncology, Daiichi Sankyo, Janssen, MSD, Novartis, Pfizer and Sanofi; and honoraria (personal) for participation in advisory boards from Arvinas, CureVac, MacroGenics and Orion.

Additional information

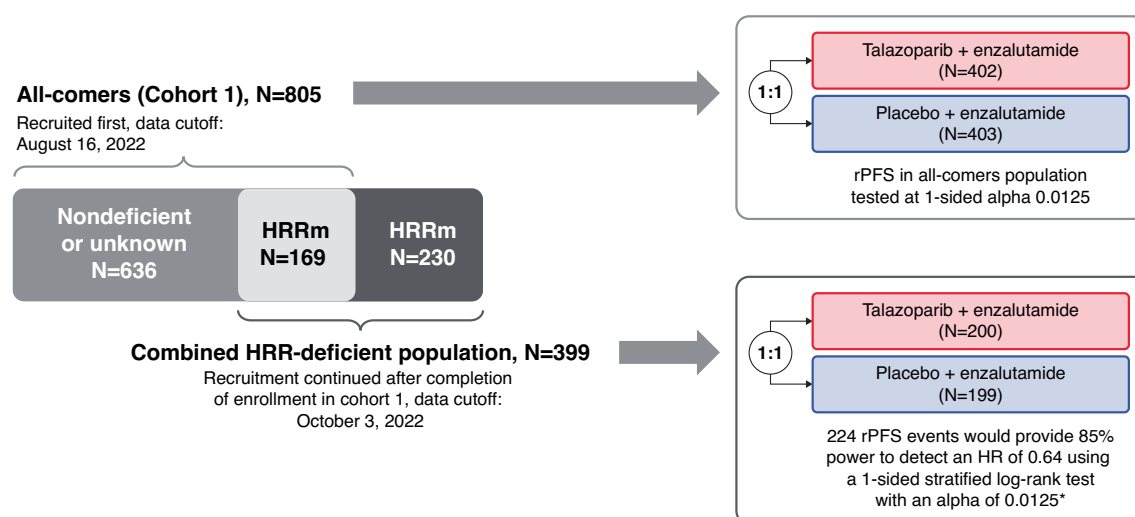
Extended data is available for this paper at <https://doi.org/10.1038/s41591-023-02704-x>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41591-023-02704-x>.

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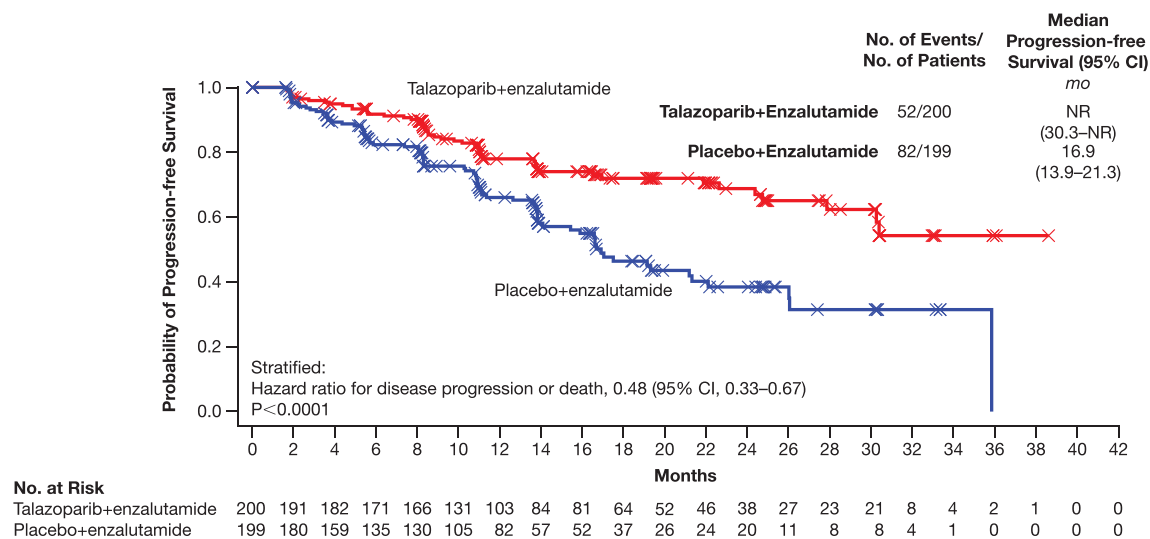
Peer review information *Nature Medicine* thanks Robert Bristow, Qian Shi, and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editor: Ulrike Harjes, in collaboration with the *Nature Medicine* team.

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Extended Data Fig. 1 | Study Cohorts and Enrollment. *An interim analysis (IA) was planned with ~70% of the total required events. The HRRm cohort would be stopped for efficacy if the pre-specified efficacy boundary was crossed ($P \leq 0.003$). As the efficacy boundary was crossed at the IA rPFS, this became the

final analysis. Survival and safety follow-up is continuing. All other endpoints are final. HRR denotes homologous recombination repair, HRRm HRR mutation-positive, and rPFS radiographic progression-free survival.



Extended Data Fig. 2 | Investigator-Assessed rPFS (HRR-Deficient Intention-to-Treat Population). rPFS was compared between treatment groups using stratified log-rank test. HRs and associated 95% two-sided CIs were estimated by a Cox proportional hazards model. Median time to event was estimated by the

Kaplan-Meier method, and 95% CIs based on the Brookmeyer-Crowley method. The P value is two-sided. CI denotes confidence interval, HR hazard ratio, HRR homologous recombination repair, NR not reached at the time of the analysis, and rPFS radiographic progression-free survival.

Extended Data Table 1 | Additional Baseline Disease Characteristics (HRR-Deficient Intention-to-Treat Population)

Characteristic	Talazoparib + Enzalutamide (N=200)	Placebo + Enzalutamide (N=199)
Initial M stage at primary diagnosis*		
M0	84 (42)	84 (42)
M1	96 (48)	95 (48)
MX	19 (10)	17 (9)
Androgen deprivation therapy at baseline		
Bilateral orchiectomy	5 (2)	11 (6)
Androgen deprivation therapy	195 (98)	187 (94)†
Tissue source for prospective HRR gene alteration testing‡		
Tumor tissue only	75 (38)	80 (40)
Tumor tissue and blood (circulating tumor DNA)	123 (62)	119 (60)
Blood (circulating tumor DNA) only	1 (<1)	0

Data are no. (%). *Not reported for the remaining patients. †Prior androgen deprivation therapy was not available in the clinical database for one patient as of the data cutoff, although this patient received androgen deprivation therapy prior to enrollment and continued to receive this treatment. ‡The tissue source was not specified for one patient in the talazoparib arm. HRR denotes homologous recombination repair.

Extended Data Table 2 | Representativeness of study participants

Category	
Disease under investigation	Metastatic castration-resistant prostate cancer
Special considerations related to:	
Age	Prevalence increases steeply with age, with the highest incidence in men >65 years of age
Race or ethnicity	African-American males have higher incidence and mortality rates for prostate cancer than White males
Geography	Throughout the world, prostate cancer incidence and mortality rates vary widely between countries, with mortality rates in 2020 between 7.8 and 13.7 per 100,000 in Europe, 8.3 in North America, and some of the lowest mortality rates in Asia (4.6 per 100,000 in East Asia)
Prior treatment	Many patients with metastatic castration-resistant prostate cancer will have received prior treatment for castration-sensitive disease with androgen deprivation therapy plus a hormonal therapy, such as abiraterone or enzalutamide, possibly in combination with docetaxel
Genetic alterations	Approximately one-quarter of advanced prostate cancers are reported to have alterations in DNA damage response genes involved either directly or indirectly in HRR, the most common being <i>BRCA2</i> (approximately 10%)
Overall representativeness of this trial	The age distribution of patients was consistent with that expected, with the majority of patients aged >65 years. The proportion of Black or African-American patients who underwent randomization was small (3%). Men were enrolled from diverse geographic locations including North America (12%), Europe/United Kingdom (48%), Asia (20% in China, Japan, and Republic of Korea), and the rest of the world (19%). More than half of the men received prior first-generation anti-androgen therapy* and approximately 30% received prior docetaxel. Only 8% of patients had received prior abiraterone. The patients were selected for HRR gene alterations; approximately 40% of these had <i>BRCA1/2</i> gene alterations

*Bicalutamide, flutamide, nilutamide, cyproterone acetate. HRR denotes homologous recombination repair.

Extended Data Table 3 | Additional secondary efficacy outcomes (hrr-deficient intention-to-treat population)

	Talazoparib + Enzalutamide (N=200)	Placebo + Enzalutamide (N=199)	Hazard Ratio	P value (Two-Sided)
Median duration of response* (95% CI) — mo	20.3 (12.2–NR)	14.8 (6.6–25.8)		
PSA response $\geq 50\%$ † — n/N (%) (95% CI)	171/198 (86) (81–91)	125/199 (63) (56–70)		<0.0001
Time to initiation of subsequent antineoplastic therapy				
Patients with use — no. (%)	44 (22)	85 (43)		
Median time to use (95% CI) — mo	NR (NR–NR)	18.8 (15.4–NR)	0.40	<0.0001
Time to first symptomatic skeletal event				
Patients with event — no. (%)	36 (18)	45 (23)		
Median time to first event (95% CI) — mo	NR (33.9–NR)	NR (32.9–NR)	0.69	0.09

Time-to-event endpoints were compared between treatment groups using stratified log-rank test. HRs and associated 95% two-sided CIs were estimated by a Cox proportional hazards model. Median time-to-event endpoints were estimated by the Kaplan–Meier method, and 95% CIs based on the Brookmeyer–Crowley method. P values are two-sided. *Only includes patients with confirmed complete response or partial response: talazoparib plus enzalutamide (N=49); placebo plus enzalutamide (N=26). †The number of patients with a baseline PSA value and at least one post-baseline PSA value: talazoparib plus enzalutamide (N=198); placebo plus enzalutamide (N=199). CI denotes confidence interval, HRR homologous recombination repair, NR not reached at the time of the analysis, and PSA prostate-specific antigen.

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|-------------------------------------|-------------------------------------|--|
| ☐ | ☒ | The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☐ | ☒ | A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
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<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ | For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
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| ☒ | ☐ | Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated |

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Software and code

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Data collection Oracle Clinical Remote Data Capture was used for data collection.

Data analysis SAS version 9.4 was used for the data analysis.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

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Reporting on sex and gender	The study was in patients with metastatic castration-resistant prostate cancer; therefore, all participants were of male sex.
Reporting on race, ethnicity, or other socially relevant groupings	The baseline characteristics, including race, are reported in Table 1. As noted in Table S2 on the representativeness of the study participants, the proportion of Black or African-American patients who underwent randomization was small (3%). Men were enrolled from diverse geographic locations including North America (12%), Europe/United Kingdom (48%), Asia (20% in China, Japan, and Republic of Korea), and the rest of the world (19%).
Population characteristics	The baseline characteristics are reported in Table 1. The age distribution of patients was consistent with that expected, with the majority of patients aged >65 years. This manuscript reports the results in the HRR-deficient population selected for HRR gene alterations; approximately 40% of these had BRCA1/2 gene alterations. More than half of the men received prior first-generation anti-androgen therapy (bicalutamide, flutamide, nilutamide, cyproterone acetate) and approximately 30% received prior docetaxel. Only 8% of patients had received prior abiraterone.
Recruitment	Patients were enrolled in the TALAPRO-2 trial at 223 sites in 26 countries, including the USA, Europe, Israel, South America, South Africa, and the Asia-Pacific region. Part 2 of the TALAPRO-2 trial, including the HRR-deficient population reported in this manuscript, was randomized, double-blind, placebo-controlled, and an independent radiology facility was used to evaluate the primary endpoint (rPFS). Principal investigators selected patients based on their clinical judgment and their ability to ensure that the patients could meet the study requirements. In addition, there was scientific acknowledgement that the participants could benefit overall from the study intervention in alignment with randomized, double-blind, placebo-controlled trial principles. In protocol Amendment No. 6, the inclusion criteria were updated to reflect that HRR gene alteration status for participants randomized in Part 2 was determined by prospective testing using FoundationOne®CDx (tissue) and/or FoundationOne®Liquid CDx. Participants were considered HRR-deficient if the participant had one or more alteration(s) in at least one of the 12 HRR genes or if there was a discordant result between the tissue and liquid results.
Ethics oversight	The trial was conducted in accordance with the International Ethical Guidelines for Biomedical Research Involving Human Subjects, Good Clinical Practice guidelines, the principles of the Declaration of Helsinki, and local laws. The protocol and amendments were approved by the institutional review board and independent ethics committee for each site. The following independent ethics committees or Institutional Review Boards provided study approval: Comité de Revision Institucional - Hospital Británico de Buenos Aires, CABA, Argentina; Comité de Ética "Dr. Claude Bernard", Rosario, Argentina; Comité de Ética en Investigación - Centro de Educación Médica e Investigaciones Clínicas "Norberto Quirno" – CEMIC, CABA, Argentina; Comité de Ética en Investigación de la Fundación OncoSalud (CEIFOS), Pergamino, Argentina; Comité Independiente De Ética Para Ensayos En Farmacología Clínica, CABA, Argentina; Comité Institucional de Ética de la Investigación en Salud (C.I.E.I.S.) de la Clínica Universitaria Reina Fabiola, Córdoba, Argentina; Comité Institucional de Ética de Investigación en Salud del Hospital Privado Centro Médico de Córdoba, Córdoba, Argentina; St Vincent's Hospital Human Research Ethics Committee, Darlinghurst, Australia; Bellberry Limited, Eastwood, Australia; Commissie Voor Medische Ethiek, Gent, Belgium; Comissão Nacional de Ética em Pesquisa/CONEP, Brasília, Brazil; Comité de Ética em Pesquisa da Fundação Pio XII - Hospital de Câncer de Barretos, Barretos, Brazil; Comité de Ética em Pesquisa da Universidade do Vale do Taquari – UNIVATES, Lajeado, Brazil; Comité de Ética em Pesquisa do Hospital Mae de Deus, Porto Alegre, Brazil; Comité de Ética em Pesquisa do Instituto D'Or de Pesquisa e Ensino, Rio de Janeiro, Brazil; Comité de Ética em Pesquisa-Hospital Universitario Pedro Ernesto, Rio de Janeiro, Brazil; Comité de Ética em Pesquisa da Universidade Regional do Noroeste do estado do Rio Grande do Sul, Ijuí, Brazil; Comité de Ética em Pesquisa - CEP do Hospital das Clínicas da Faculdade de Medicina da Universidade de São Paulo - HCFMUSP, São Paulo, Brazil; Comité de Ética em Pesquisa da Faculdade de Medicina do ABC, Santo André, Brazil; Comitê de Ética em Pesquisa do Instituto Nacional de Câncer José Alencar Gomes da Silva – INCA, Rio de Janeiro, Brazil; Comité de Ética em Pesquisa do Hospital Nossa Senhora da Conceição - Grupo Hospitalar Conceição, Porto Alegre, Brazil; Comité de Ética em Pesquisa da Sociedade Beneficente de Senhoras Hospital Sirio Libanes, São Paulo, Brazil; Comitê de Ética em Pesquisa do Hospital Alemão Oswaldo Cruz – SP, São Paulo, Brazil; Comité de Ética em Pesquisa da Pontifícia Universidade Católica do Rio Grande do Sul-PUC/RS, Porto Alegre, Brazil; Comité d'éthique de la recherche du CHUM, Montreal, Canada; Health Research Ethics Board of Alberta - Cancer Committee, Edmonton, Canada; Ontario Cancer Research Ethics Board, Toronto, Canada; Comité de Ética Científico Servicio de Salud Metropolitano Oriente, Santiago, Chile; Comité Ético Científico Hospital Dr. Gustavo Fricke Servicio de Salud Vina del Mar – Quillota, Vina del Mar, Chile; Comité de Ética Científica Servicio Salud Araucanía Sur, Temuco, Chile; Ethics Committee of Zhongshan Hospital Fudan University, Shanghai, China; Ethics Committee of The First Affiliated Hospital of Xi'an Jiaotong University, Xi'an, China; Ethics committee of Zhejiang Cancer Hospital, Hangzhou, China; Ethics Committee of National Cancer Center/ Cancer Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China; Ethics Committee of Chongqing University Cancer Hospital, Chongqing, China; Ethics Committee of Beijing Cancer Hospital, Beijing, China; Ethics Committee of The First Affiliated Hospital of Anhui Medical University, Hefei, China; Medical Ethics Committee of First Affiliated Hospital of Xiamen University, Xiamen, China; Wuxi People's Hospital Ethics Committee, Wuxi, China; Ethics Committee of Nanjing Drum Tower Hospital, Nanjing, China; Ethics Committee of Shanghai Tenth People's Hospital, Shanghai, China; Clinical Trial Ethics Committee of Huazhong University of Science and Technology, Wuhan, China; Ethics Committee of Huashan Hospital, Fudan University, Shanghai, China; Ethics Committee of Fudan University Cancer Hospital, Shanghai, China; Ethics Committee of Beijing Hospital, Beijing, China; Ethics Committee of Peking University First Hospital, Beijing, China; Peking University Third Hospital Medical Science Research Ethics Committee, Beijing, China; Ethics Committee for Clinical Trials of Drugs (Medical Apparatus) of Ningbo First Hospital, Ningbo, China; Ethics Committee of Ruijin Hospital Affiliated to Shanghai Jiaotong University School of Medicine, Shanghai, China; West China Hospital of Sichuan University Clinical Trial Ethics Committee, Chengdu, China; EC of Second Affiliated Hospital of Suzhou University, Suzhou, China; Shanghai General Hospital Medical Ethics Committee, Shanghai, China; Ethics Committee of Nanjing First Hospital, Nanjing, China; Clinical Trial Ethics Committee of Huazhong University of Science and Technology, Wuhan, China; Drug and Machinery Clinical trial Branch of EC of The First Affil. 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Helsingin ja Uudenmaan sairaanhoitopiiri, Helsinki, Finland; Comite De Protection Des Personnes (CPP) Sud-Ouest Et Outre-Mer III, Bordeaux, France; Ethikkommission der Aerztekammer Hamburg, Hamburg, Germany; Egészségügyi Tudományos Tanács Klinikai Farmakológiai Etikai Bizottsága, Budapest, Hungary; Bnai Zion Medical Center Helsinki Committee, Haifa, Israel; Rambam Health Care Campus Helsinki Committee, Haifa, Israel; Tel Aviv Sourasky Medical Center Helsinki Committee, Tel Aviv, Israel; Rabin Medical Center Helsinki Committee, Petah Tikva, Israel; Shaare Zedek Medical Center Helsinki Committee, Jerusalem, Israel; Comitato Etico Azienda Ospedaliero Universitaria San Luigi Gonzaga, Orbassano, Italy; Comitato Etico Val Padana, Cremona, Italy; Comitato Etico Regionale (CER) dell'Umbria, Perugia, Italy; Comitato Etico Cardarelli-Santobono, Napoli, Italy; Comitato Etico Per Le Sperimentazioni Cliniche Dell'Azienda Provinciale Per I Servizi Sanitari, Trento, Italy; Comitato Etico della Romagna (CEROM), Meldola, Italy; Comitato Etico di Brescia, Brescia, Italy; Comitato Etico di Area Vasta Emilia Centro, Bologna, Italy; Comitato Etico IRCCS Pascale, Napoli, Italy; National Hospital Organization Central Review Board, Meguro-ku, Tokyo, Japan; National Cancer Center IRB, Chuo-ku, Tokyo, Japan; Kindai University Hospital Institutional Review Board, Osakasayama, Japan; Yokohama City University Medical Center IRB, Yokohama, Japan; Keio University Hospital Institutional Review Board, Shinjuku-ku, Tokyo, Japan; Nagoya University Hospital IRB, Nagoya, Japan; Hokkaido University Hospital Institutional Review Board, Sapporo, Japan; Tokushima University Hospital Institutional Review Board, Tokushima, Japan; Chiba Cancer Center Institutional Review Board, Chiba, Japan; Hirosaki University School of Medicine & Hospital IRB, Hirosaki, Japan; Yamagata Prefectural Central Hospital Institutional Review Board, Yamagata, Japan; Yokosuka Kyosai Hospital Institutional Review Board, Yokosuka, Japan; Hamamatsu University School of Medicine, University hospital Institutional Review Board, Hamamatsu, Japan; 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Health and Disability Ethics Committee, Wellington, New Zealand; REK Sor-Ost, Oslo, Norway; Comite Institucional de Etica en Investigacion del INEN, Lima, Peru; Comite Institucional de Bioetica de Via Libre, Lima, Peru; Komisja Bioetyczna przy Okregowej Izbie Lekarskiej w Gdansk, Gdansk, Poland; Comissao de Etica para a Investigacao Clinica, Lisboa, Portugal; University of the Witwatersrand Human Research Ethics Committee (Medical), Johannesburg, South Africa; CEIm del Hospital Universitari Vall d'Hebron, Barcelona, Spain; Etikprovningssmyndigheten, Uppsala, Sweden; Health and Care Research Wales, Wales REC 3, Cardiff, United Kingdom; Advarra IRB, Columbia, MD, United States; Vanderbilt Human Research Protection Program (VHRPP) IRB, Nashville, TN, United States; University of Utah Institutional Review Board, Salt Lake City, UT, United States; Biomedical Research Alliance of New York, LLC / Institutional Review Board, Lake Success, NY, United States; Western Institutional Review Board, Puyallup, WA, United States; Sharp HealthCare Institutional Review Board, San Diego, CA, United States; Schulman Associates Institutional Review Board, Cincinnati, OH, United States; Loma Linda University Health - Institutional Review Board, Loma Linda, CA, United States; Administrative Panels on Human Subjects in Medical Research ("Stanford IRB"), Palo Alto, CA, United States; University of Maryland, Baltimore - Institutional Review Board, Baltimore, MD, United States; Cook County Health Office of Research and Regulatory Affairs, Chicago, IL, United States; Samaritan Health Services Regional Institutional Review Board, Corvallis, OR, United States; University of Iowa IRB-01, Human Subjects Office, Iowa City, IA, United States; Lakeland Regional Medical Center, Inc. IRB, Lakeland, FL, United States; VA Med Ctr, Long Beach CA IRB #1, Long Beach, CA, United States; Rush University Medical Center Institutional Review Board, Chicago, IL, United States; UCLA Office of the Human Research Protection Program, Los Angeles, CA, United States; VA Saint Louis Healthcare System Institutional Review Board, St. Louis, MO, United States; Baylor Scott and White Research Institute Institutional Review Board-Gold, Temple, TX, United States; Providence St. Joseph Health IRB, Renton, WA, United States; IntegReview, Austin, TX, United States; Kaiser Permanente Northwest Institutional Review Board, Portland, OR, United States; Ochsner Institutional Review Board, New Orleans, LA, United States; Eisenhower Medical Center, Institutional Review Board, Rancho Mirage, CA, United States. The full list of TALAPRO-2 investigators is included in the Supplementary Information table. All patients provided written informed consent.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

Sample size and power calculation were based on the log-rank test. For the primary comparison in the HRR-deficient population, 224

Sample size	progression-free survival events would provide 85% power to detect a hazard ratio of 0.64 using a one-sided stratified log-rank test at a significance level of 0.0125 with two pre-specified interim analyses based on Lan-DeMets α -spending and β -spending functions. Approximately 380 patients with HRR gene alterations were to be enrolled.
Data exclusions	Inclusion criteria for the TALAPRO-2 trial were pre-specified and previously published (Agarwal N et al. Future Oncol 2022;18(4):425-436).
Replication	Part 2 of TALAPRO-2 was a randomized, double-blind, placebo-controlled trial including a large number (N=399) of patients with HRR-deficient mCRPC. The FoundationOne®CDx and/or FoundationOne®Liquid CDx next-generation sequencing test was used for prospective assessment of HRR gene alterations. Enrollment of patients with ATM and/or CDK12 gene alterations was paused between January–November 2021 as their observed prevalence exceeded expectations and was anticipated to suppress representation of alterations in the remaining genes under study. The pause in enrollment of patients with ATM and/or CDK12 gene alterations was driven by expected prevalence numbers based on the largest and most comprehensive prospective assessment of prostate cancer tumor samples using the FoundationOne® Assay. This pause occurred in a blinded fashion regarding distribution of HRR alterations to the two treatment arms and allowed a rebalancing of the distribution across the 12 gene panel in an effort to best reflect the prevalence in mCRPC.
Randomization	Participants were randomized 1:1 by site personnel using a centralized Interactive Web Response System to talazoparib plus enzalutamide or matching placebo plus enzalutamide. Randomization was stratified by prior novel hormonal therapy or docetaxel for castration-sensitive prostate cancer.
Blinding	The sponsor, participants, and investigators were blinded to talazoparib or placebo during data collection, while enzalutamide was open-label (both treatment groups received enzalutamide). The blinding methodology in TALAPRO-2 prevented selection or ascertainment (i.e., information) biases and potentially improved the compliance and the retention of trial participants. In TALAPRO-2 the withholding of information about the assigned interventions from participants involved in the trial played a potentially major role in mitigating threats to the internal validity of the trial outcomes.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☐	☒ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Clinical data

Policy information about [clinical studies](#)

All manuscripts should comply with the ICMJE [guidelines for publication of clinical research](#) and a completed [CONSORT checklist](#) must be included with all submissions.

Clinical trial registration	NCT03395197
Study protocol	A redacted version of the protocol will be available as part of the Supplementary Information.
Data collection	Data collection occurred at each study site (study start date: Dec 18, 2017; primary completion date: Oct 3, 2022 [NCT03395197; ClinicalTrials.gov]). Patients from 223 sites in 26 countries, including the USA, Europe, Israel, South America, South Africa, and the Asia-Pacific region were enrolled in the TALAPRO-2 trial (Agarwal N et al. Future Oncol 2022;18(4):425-436).
Outcomes	The primary endpoint was rPFS by blinded independent central review per RECIST 1.1 and PCWG3. Key secondary endpoints were overall survival, objective response rate, duration of soft tissue response, time to PSA progression, PSA response, PFS2 (investigator-assessed), safety, patient-reported outcomes, and pharmacokinetics.